0.1 Minimal Polynomial

Definition 0.1.1. Let V be a Vector Space over F, and let $T: V \to V$ be a linear operator. A map

$$\varphi_T : F[x] \to \mathcal{L}(V) : p(x) \mapsto p(T)$$

is called *evaluation* of T. It is a ring-homomorphism, and $\ker \varphi_T$ is principal ideal. Moreover,

$$F[x]/\ker \varphi_T \cong F[T]$$

by the first-isomorphism theorem.

Note: $\mathcal{L}(V) = \{T : V \to V \mid T$ is linear operator$\}$ has algebraic structures:

$(\mathcal{L}(V), +, \cdot)$ is Vector Space over F and $(\mathcal{L}(V), +, \circ)$ is a Ring with identity.

Generally, $(\mathcal{L}(V), +, \circ)$ is non-commutative, and <u>Not</u> domain. For instance, let $V = F^n$. Then, for each $1 \le j \le n$,

$$A_j : F^n \to F^n : (a_i)_{i=1}^n \mapsto (0, \ldots, a_j, 0, \ldots)$$

are not zero maps, but $E_1 \circ E_2 = \mathbf{0}$. And ask a toddler on the street, examples for Non-Commutative.
But, the subring

$$F[T] = \{p(T) \mid p(x) \in F[x]\} \subset \mathcal{L}(V)$$

is Commutative Ring with identity, still it is not a domain, in general. Moreover, as a Subspace,

$$F[T] = \mathrm{span}_F\{T^n \mid n = 0, 1, \ldots\}$$

If V is finite-dimensional space, then $\mathcal{L}(V)$ has a n^2 dimension, therefore

$$T^0, T^1, \ldots, T^{n^2}$$

is linearly dependent, this implies there exists a polynomial $p(x) \in F[x]$ such that $p(T) = 0$. In other words,

If V has a finite dimension, then for linear operator $T : V \to V$, $\ker \varphi_T$ is non-zero principal ideal.

In this case, the unique monic polynomial of $\ker \varphi_T$ is called the *minimal polynomial* for T.

Thm. The charaterstic and minimal polynomials for T have the same roots.

Proof. Let p be the minimal polynomial for T.

Suppose that $c \in F$ is root of p. That is, $p(c) = 0$. Then,

$$p = (x - c) \cdot q \quad (\deg q < \deg p)$$

for some $q \in F[x]$. Since p is the minimal, $q(T) \neq 0$. Take $\beta \in V$ s.t. $q(T)(\beta) \neq 0$.
Then, $0 = p(T)(\beta) = (T - cI) \cdot q(T)(\beta)$. Thus $T \circ q(T)(\beta) = c\beta$, c is eigenvalue.
Conversely, let $c \in F$ be an eigenvalue of T. That is, $T\beta = c\beta$ for some nonzero $\beta \in V$.
Now, $0 = p(T)(\beta) = p(c)\beta$ and since β is nonzero, $p(c) = 0$. ∎

「Conductor.

「Def. Let W be an invariant subspace for $T: V \to V$, and $\alpha \in V$ be a vector. Define T-Conductor of α into W as;

$$S_T(\alpha; W) \overset{def}{=} \{ g \in F[x] \mid g(T)(\alpha) \in W \}.$$

In particular, if $W = \{0\}$ the Conductor is called the T-annihilator of α.

「Lemma. The Conductor $S(\alpha; W)$ is an Ideal in $F[x]$.

Proof. First, if $T[W] \subseteq W$, then $g(T)[W] \subseteq W$, because for any $n \in \mathbb{N}$,

$$\beta \in W \Rightarrow T^n(\beta) = T^{n-1}(\underbrace{T(\beta)}_{\in W}) = \cdots = T(T^{n-1}(\beta)) \in W.$$

And since $W \leq V$, it is closed under the addition and Scalar multiplication, $g(T)[W] \subseteq W$. Now, to show $S(\alpha; W)$ is an Ideal, first show that it is space. For any $f, g \in S(\alpha; W)$ and $c \in F$, $(cf+g)(T) = cf(T) + g(T)$, theref

$$(cf+g)(T)(\alpha) = cf(T)(\alpha) + g(T)(\alpha) \in W$$

Thus it is subspace. Moreover, for any $f \in F[x]$ and any $g \in S(\alpha; T)$,

$$(fg)(T)(\alpha) = \underbrace{f(T)\underbrace{g(T)(\alpha)}_{\in W}}_{\in W} \in W.$$

Thus $fg \in S(\alpha; W)$. ■

Note: Given the linear operator T and T-invariant subspace W, and any vector $\alpha \in V$,

$$\varphi: F[x] \to V/W : f(x) \mapsto f(T)(\alpha) + W$$

is $F[x]$-module Homomorphism, so $S_T(\alpha; W) = \ker \varphi$ is principal Ideal in $F[x]$.

Lemma. Let V be a fin. dim v.s over F, and let $T: V \to V$ be a linear operator. Suppose that the minimal polynomial of T splits over F as:

$$p = (x-c_1)^{r_1} \cdot \dots \cdot (x-c_k)^{r_k} \quad (c_i \in F)$$

Then, for any proper T-invariant subspace $W < V$, there exists a vector $\alpha \in V$ s.t

$\alpha \notin W$ and for some c_i, $(T - c_i I)(\alpha) \in W$.

Proof. Let $\beta \in V \setminus W$ be given, and let $g \in F[x]$ be a T-conductor of β into W. Then, g divides p. And, since $\beta \notin W$, g is not constant. Therefore,

$$g = (x-c_1)^{e_1} \cdots (x-c_k)^{e_k} \quad (0 \le e_i \le r_i)$$

and at least one of e_i is positive. Put j s.t $e_j > 0$. Then,

$$g = (x-c_j) \cdot h \quad \text{where } h = g/(x-c_j) \in F[x].$$

Since g is T-conductor of β, $h(T)(\beta) \notin W$, but $(T-c_j I)(h(T)(\beta)) = g(T)(\beta) \in W$.

Thm. Let V be a finite dimensional vector space over F, and $T: V \to V$ be a linear. T is Triangulable if and only if minimal polynomial of T splits over F.

Proof. If T is triangulable with respects to a basis β, then the characteristic polynomial is

$$f(x) = \det([T]_\beta - xI) = \prod_{i=1}^{n} (A_{ii} - x) \quad \text{where } A_{ii} \text{ is } i\text{'th diagonal element of } [T]_\beta$$

thus, Proved. Conversely, choose $\alpha_1 \in V$ s.t $\alpha_1 \notin \{0\}$, $(T - c_1 I)(\alpha_1) \in \{0\}$ for some eigenvalue $c_1 \in F$ of T. Choose $\alpha_2 \in V$ s.t $\alpha_2 \notin \langle \alpha_1 \rangle$, $(T - c_2 I)(\alpha_2) \in \langle \alpha_1 \rangle$, it is possible because $\langle \alpha_1 \rangle$ is T-invariant. Again, choose $\alpha_3 \in V$ s.t $\alpha_3 \notin \langle \alpha_1, \alpha_2 \rangle$, $(T - c_3 I)(\alpha_3) \in \langle \alpha_1, \alpha_2 \rangle$. It is possible again, $T(\alpha_2) \in \langle \alpha_1, \alpha_2 \rangle$.
Now, we obtain $\{\alpha_1, \dots, \alpha_n\}$ s.t

$$T(\alpha_1) = c_1 \alpha_1$$
$$T(\alpha_2) = a_{12}\alpha_1 + c_2 \alpha_2$$
$$\vdots$$
$$T(\alpha_n) = a_{1n}\alpha_1 + a_{2n}\alpha_2 + \dots + c_n \cdot \alpha_n$$

「Thm. Let V be a fin. dim. v.s over F, and $T: V \to V$ be a linear operator.
T is diagonalizable if and only if minimal polynomial of T splits and separable.

Proof. Suppose that T is diagonalizable. Let $c_1, \cdots, c_k$ be distinct eigenvalues.
Then, for any eigenvector $\alpha \in V$, there is a $c_i \in F$ s.t $(T - c_i I)(\alpha) = 0$.
Therefore, for $p = (x - c_1) \cdots (x - c_k)$,

$$p(T) = (T - c_1 I) \cdots (T - c_k I)(\alpha) = 0$$

because $\{T - c_1 I, \cdots, T - c_k I\}$ commute. Since T is diagonalizable,
for any vector of V can be represented by linear combination of eigenvectors,

$$p(T) = 0$$

Thus proved. Conversely, let W be a spanned subspace by all of eigenvectors of T.
Suppose that $W \neq V$. By the construction, W is T-invariant. Therefore, by the Lemma,
there is $\alpha \in V \setminus W$ and eigenvalue $c_j \in F$ s.t $\beta = (T - c_j I)(\alpha) \in W$.
Since $\beta \in W$, $\beta = \beta_1 + \cdots + \beta_k$ where $T(\beta_i) = c_i \beta_i$, $1 \le i \le k$. Therefore, for any $h \in F[x]$

$$h(T)(\beta) = \sum_{i=1}^{k} h(c_i) \cdot \beta_i \in W.$$

Put $p = (x - c_j) g$. Also, $g - g(c_j) = (x - c_j) h$. Then,

$$g(T)(\alpha) - g(c_j)(\alpha) = h(T)(T - c_j I)(\alpha) = h(T)(\beta) \in W$$

And, $0 = p(T)(\alpha) = (T - c_j I) g(T)(\alpha) = g(T)(T - c_j I)(\alpha) \in W$.
Therefore $g(c_j)(\alpha) \in W$, and since $\alpha \notin W$, $g(c_j) = 0$. Contradicts to $p = (x - c_j)^2 \cdot g'$.

「Summary.

Let $T : V \to V$ be a linear operator on the fin.dim. V over F.
For the characteristic polynomial $f(t) \in F[t]$, and the minimal polynomial $P(t) \in F[t]$, $P(t)$ divides $f(t)$ by the Caley-Hamilton Thm, and for any T-invariant $W < V$ and $\alpha \in V$, every T-conductor of α into W divides $P(t)$. Moreover, $P(t)$ and $f(t)$ share the roots, so,

「T is triangularizable $\Leftrightarrow$ $P(t)$ splits over F $\xLeftrightarrow[\text{same root}]{P|f}$ $f(t)$ splits over F $\Leftrightarrow$ T has a Jordan form.

「$P(t)$ splits over F and separable $\Leftrightarrow$ T is diagonalizable

$\not\Rightarrow$ $f(t)$ is separable.

「Anyway, T has a Primary decomposition, moreover, T has a Rational-Normal form.」

「Calculation.

Let $D : P_n(\mathbb{R}) \to P_n(\mathbb{R}) : P(x) \mapsto D_x P(x)$.

Find the minimal polynomial of D and xD, and find the Jordan form.

Proof. Clearly, for all x^m $(0 \leq m \leq n)$,

$$D^{n+1}(x^m) = 0, \quad \text{so} \quad D^{n+1} = 0.$$

And, $\{I, D, \cdots, D^n\}$ is linearly independent, because:

Suppose that $c_0 I + c_1 D + \cdots + c_n D^n = 0$.

Substituting $\frac{1}{m!} x^m$ for each $m = 0, 1, \cdots, n$, then

$$0 = (c_0 I + \cdots + c_n D^n)\left(\frac{1}{0!} \cdot x^0\right) = c_0$$

$$0 = (c_1 D + \cdots + c_n D^n)\left(\frac{1}{1!} \cdot x\right) = c_1$$

$$\vdots$$

So, $c_0 = \cdots = c_n = 0$. Therefore, the minimal polynomial of D is t^{n+1}.

Meanwhile,
$$xD(1) = 0,$$
$$xD(x) = x,$$
$$xD(x^2) = 2x^2,$$
$$\text{"} \quad 3x^3,$$
$$\vdots$$

Therefore, $[xD]_\beta = \begin{bmatrix} 0 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & & \vdots \\ \vdots & \vdots & 2 & & \vdots \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & n \end{bmatrix}$, where $\beta = \{1, \cdots, x^n\}$ is the standard basis for $P_n(\mathbb{R})$.

So, the characteristic polynomial is $t \cdot (t-1) \cdots (t-n)$, distinct $n+1$ roots, the minimal polynomial is same. Now, the Jordan-Normal form are

$$[D] = \begin{bmatrix} 0 & 1 & & 0 \\ & 0 & 1 & \\ & & 0 & \ddots \\ 0 & & & 0 \end{bmatrix}, \quad [xD] = \begin{bmatrix} 0 & 0 & 0 & \cdots & \vdots \\ \vdots & 1 & 0 & & \vdots \\ \vdots & 0 & 2 & \cdots & \vdots \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & 0 & & n \end{bmatrix} \text{ (diagonal).}$$

「Jordan and Rational form of 3×3 Matrix.

$$A=\begin{pmatrix} 0 & -3 & 7 \\ -1 & -1 & 5 \\ -1 & -2 & 6 \end{pmatrix}$$

t^2-4t+4

$$f(t)=\det(-A+tI)=\begin{vmatrix} t & 3 & -7 \\ 1 & t+1 & -5 \\ 1 & 2 & t-6 \end{vmatrix}=t\cdot\left[(t+1)(t-6)+10\right]-(3t-18+14)+(-15+7t+7)$$

$$=t(t^2-5t+4)-(3t-4)+(7t-8)$$

$$=t^3-5t^2+4t-3t+4+7t-8$$

$$=t^3-5t^2+8t-4$$

$$=(t-1)(t-2)^2.$$

Clearly, $A-I\neq 0$ and $A-2I\neq 0$.

By the Caley-Hamiton, $f(A)=0$. So, we will check that either

$(A-I)(A-2I)$ or $\cancel{(A-2I)^2}$ ← has same root, so not neccesary. is zero.

1). $(A-I)(A-2I)=\begin{pmatrix} -1 & -3 & 7 \\ -1 & -2 & 5 \\ -1 & -2 & 5 \end{pmatrix}\cdot\begin{pmatrix} -2 & -3 & 7 \\ -1 & -1 & 5 \\ -1 & -2 & 6 \end{pmatrix}=\begin{pmatrix} 2+3-7 & & \\ & \ddots & \\ & & \end{pmatrix}\neq 0.$

So, $f(t)$ is the characteristic polynomial and the minimal polynomial.

Now, $f(t)$ is not separable but splits, so A has the Jordan form $\begin{bmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

And has a rational form $\begin{bmatrix} 0 & 0 & 4 \\ 1 & 0 & -8 \\ 0 & 1 & 5 \end{bmatrix}$

Rational form Exercise.

Let $A=\begin{bmatrix}5&-6&-6\\-1&4&2\\3&-6&-4\end{bmatrix}$

$$\det(tI-A)=\begin{vmatrix}t-5&6&6\\1&t-4&-2\\-3&6&t+4\end{vmatrix}
\begin{aligned}&=(t-5)(t^2-16+12)-(6t+24-36)-3(-12-6t+24)\\
&=(t-5)(t^2-4)-(6t-12)+3(6t-12)\\
&=t^3-4t-5t^2+20+12t-24\\
&=t^3-5t^2+8t-4\\
&=(t-1)(t-2)^2.\end{aligned}$$

The possible: $(t-1)(t-2)^i$ $(i=1,2)$.

$$(A-I)(A-2I)=\begin{pmatrix}4&-6&-6\\-1&3&2\\3&-6&-5\end{pmatrix}\cdot\begin{pmatrix}3&-6&-6\\-1&2&2\\3&-6&-6\end{pmatrix}=\begin{pmatrix}&&\\&0&\\&&\end{pmatrix}$$

So, $A\sim\begin{bmatrix}A_1&0\\0&A_2\end{bmatrix}$ where $A_1=\begin{bmatrix}0&-2\\1&3\end{bmatrix}$, $A_2=[2]$.